

## **PROJECT REPORT**

### **Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer**

Platform: ServiceNow (Flow Designer)

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## **1. INTRODUCTION**

### **1.1 Project Overview**

This project automates the ordering and configuration of standard laptops in an IT procurement process using the Flow Designer of ServiceNow. In the existing process, a request for a standard laptop must be followed by a manual configuration task, which is often missed or delayed. The project builds a flow that is triggered from the Service Catalog. Once the service request is approved, the flow automatically creates a Catalog Task with the short description "Laptop need to Configured" and assigns it to the Hardware group.

The flow is then attached to the "Standard Laptop" catalog item through the Process Engine, so every new order runs the automation without manual intervention.

### **1.2 Purpose**

The purpose of this project is to reduce delays and manual overhead in standard laptop procurement. The project aims to:

- Create a seamless experience for users requesting standard laptops by ensuring timely configuration.
- Reduce manual intervention and potential errors in the procurement process.
- Improve resource utilisation within the IT department by optimising task allocation.
- Enhance overall efficiency and productivity in IT procurement operations.

## **2. IDEATION PHASE**

The current IT procurement process lacks efficiency and automation, resulting in delays and manual overhead, particularly in handling standard laptop orders. Requests for standard laptops often require configuration, but this step is prone to oversight or delay, leading to frustration among users and inefficient resource allocation within the IT department.

User Story: As a member of the IT procurement team, I want a streamlined process for ordering standard laptops so that tasks are automatically generated and assigned to the hardware team for configuration. This will ensure that laptops are promptly configured upon arrival, reducing user wait times and minimising manual intervention.

### **2.2 Empathy Map Canvas**

| Quadrant    | Details                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------|
| Says        | "My laptop request is taking too long."<br>"Has the laptop been configured yet?"                 |
| Thinks      | Manual follow-ups waste time.<br>The configuration step may be forgotten.                        |
| Does        | Raises a request in the Service Catalog.<br>Follows up repeatedly with IT and the Hardware team. |
| Feels       | Frustrated by delays and lack of visibility.<br>Relieved when tasks are assigned automatically.  |
| Pain Points | Missed or delayed configuration tasks, manual effort, unclear ownership.                         |
| Gains       | Automatic task creation, faster laptop delivery, less manual work for the IT team.               |

## 2.3 Brainstorming

The team discussed several ways to remove the manual step in laptop configuration:

- Manually reminding the Hardware team by e-mail or chat – rejected, still manual and error-prone.
- Creating a business rule with scripting – possible, but harder to maintain for non-developers.
- Using Flow Designer with a Service Catalog trigger – selected, because it is a low-code, easy to maintain and easy to extend approach.

Final idea: trigger a flow on the Service Catalog request, create a Catalog Task after approval, and assign it automatically to the Hardware group.

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

| Stage         | User Action                                                             | System Response                                        | Experience       |
|---------------|-------------------------------------------------------------------------|--------------------------------------------------------|------------------|
| Request       | Opens Service Catalog > Hardware > Standard Laptop and clicks Order Now | Request and Requested Item are created                 | Simple and quick |
| Approval      | Approver opens the request and approves it                              | Approval status changes to Approved                    | Clear            |
| Automation    | No action needed                                                        | Flow creates a Catalog Task and assigns it to Hardware | Effortless       |
| Configuration | Hardware team sees the task "Laptop need to Configured"                 | Task is tracked against the Requested Item             | Timely           |
| Delivery      | Receives the configured laptop                                          | Request is closed                                      | Satisfied        |

### 3.2 Solution Requirement

Functional requirements:

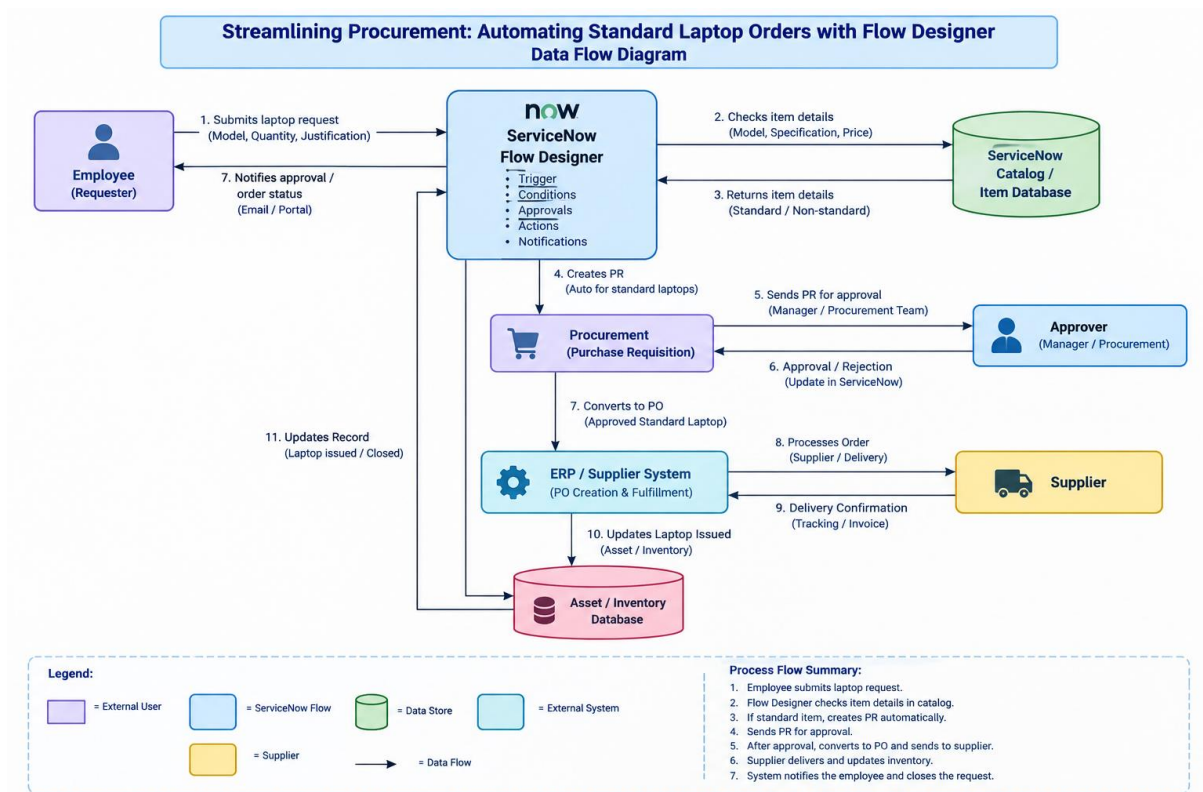
- Service Catalog item "Standard Laptop" available under the Hardware category.
- A flow named "Standard laptop task" with a Service Catalog trigger.
- An action to create a Catalog Task linked to the Requested Item.
- Task fields: short description "Laptop need to Configured", assignment group "Hardware", approval "Approved".
- Flow attached to the catalog item through the Process Engine.

Non-functional requirements: reliability, easy maintenance, role-based access and traceability of every request.

### 3.3 Data Flow Diagram

| Step | From | Process | To |
|------|------|---------|----|
|------|------|---------|----|

|   |                     |                                                        |                            |
|---|---------------------|--------------------------------------------------------|----------------------------|
| 1 | User                | Orders Standard Laptop from Service Catalog            | Request / Requested Item   |
| 2 | Approver            | Approves the request                                   | Approval record            |
| 3 | Flow Designer       | Service Catalog trigger fires after approval           | Create Catalog Task action |
| 4 | Create Catalog Task | Creates task with short description and Hardware group | Catalog Task table         |
| 5 | Hardware team       | Configures the laptop and updates the task             | Requested Item             |



### 3.4 Technology Stack

| Component       | Technology                                                      |
|-----------------|-----------------------------------------------------------------|
| Platform        | ServiceNow (Personal Developer Instance)                        |
| Automation tool | Flow Designer (Process Automation)                              |
| Trigger         | Service Catalog                                                 |
| Action          | Create Catalog Task                                             |
| Modules used    | Service Catalog, Maintain Items, Requested Items, Catalog Tasks |
| Access          | Web browser                                                     |

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The problem is a delayed and manual laptop configuration step. The solution is an automated flow that creates and assigns the configuration task as soon as the request is approved. This fits the need of the IT procurement team (less manual work) and the end user (faster laptop readiness).

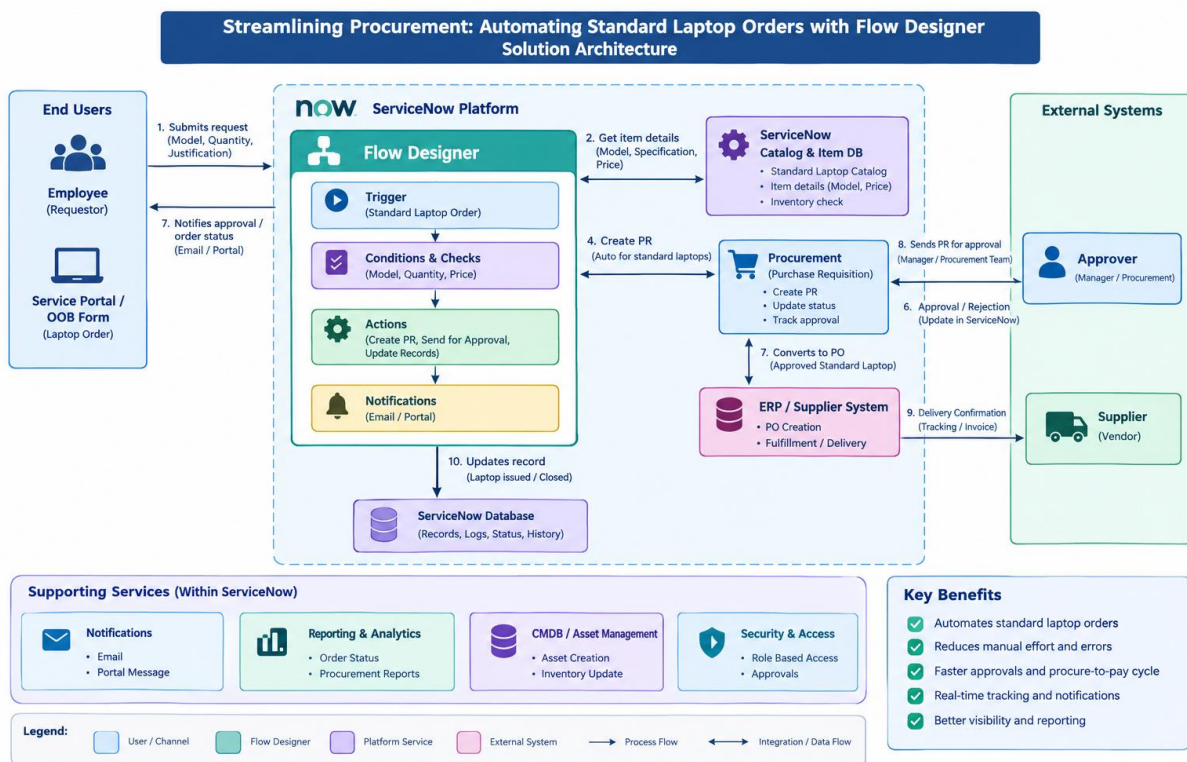
### 4.2 Proposed Solution

A flow named "Standard laptop task" is created in Flow Designer with the Service Catalog trigger. The Create Catalog Task action uses the Requested Item record and fills in the short description, description, assignment group (Hardware) and approval (Approved). The flow is activated and mapped to the Standard Laptop catalog item under the Process Engine, replacing other automations.

### 4.3 Solution Architecture

| Layer      | Description                                                     |
|------------|-----------------------------------------------------------------|
| User layer | Employee orders the Standard Laptop through the Service Catalog |

|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| Approval layer   | Approver approves the request from the Approvers section            |
| Automation layer | Flow Designer: Service Catalog trigger > Create Catalog Task action |
| Data layer       | Request, Requested Item and Catalog Task tables                     |
| Fulfilment layer | Hardware assignment group configures the laptop                     |



## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Milestone | Activity | Status |
|-----------|----------|--------|
|-----------|----------|--------|

|                              |                                                                                              |           |
|------------------------------|----------------------------------------------------------------------------------------------|-----------|
| Milestone 1: Flow            | Create a flow with Service Catalog trigger and Create Catalog Task action; save and activate | Completed |
| Milestone 2: Flow assignment | Attach the flow to the Standard Laptop item using the Process Engine                         | Completed |
| Milestone 3: Service Catalog | Place the order, approve it and verify the Catalog Task                                      | Completed |

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

| Test Case             | Expected Result                                                 | Result |
|-----------------------|-----------------------------------------------------------------|--------|
| Order Standard Laptop | Request and Requested Item are created                          | Passed |
| Approve the request   | Approval status becomes Approved                                | Passed |
| Flow execution        | Flow runs automatically after approval                          | Passed |
| Catalog Task creation | Task created with short description "Laptop need to Configured" | Passed |
| Task assignment       | Assignment group is Hardware                                    | Passed |

## 7. RESULTS

### 7.1 Output Screenshots

#### Milestone 1 – Creating the flow (Standard laptop task)

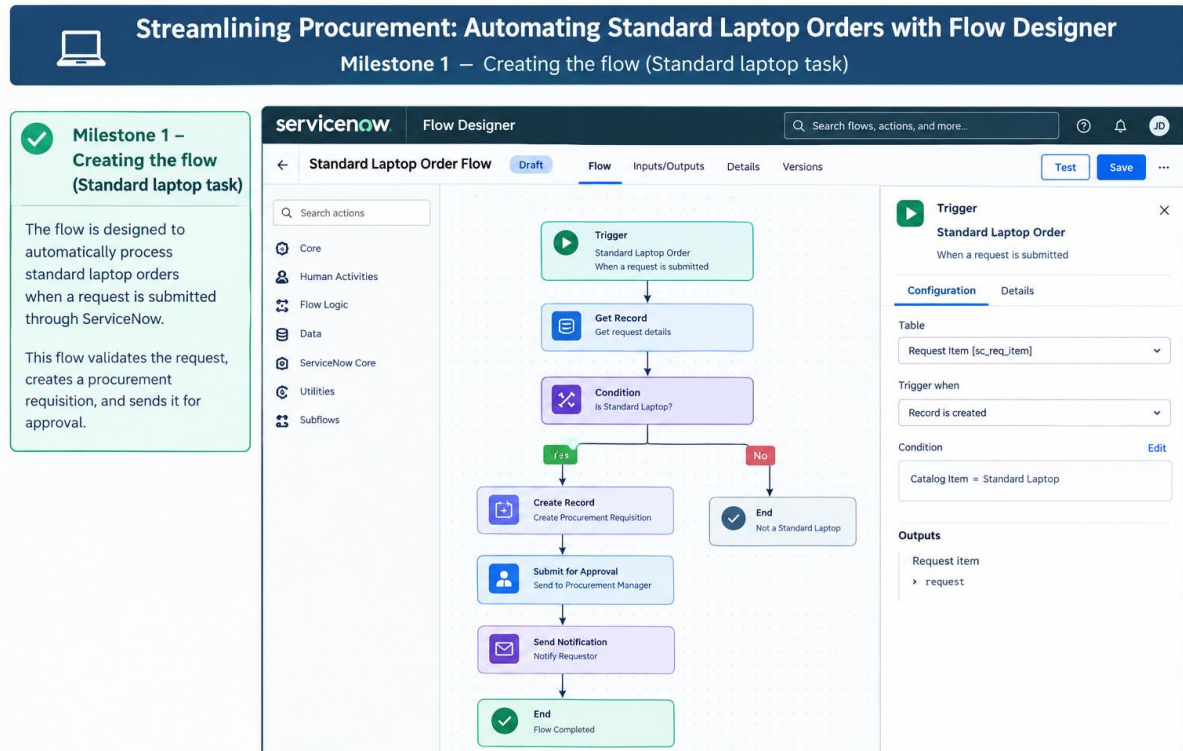


Figure 1: Opening Flow Designer from the All menu

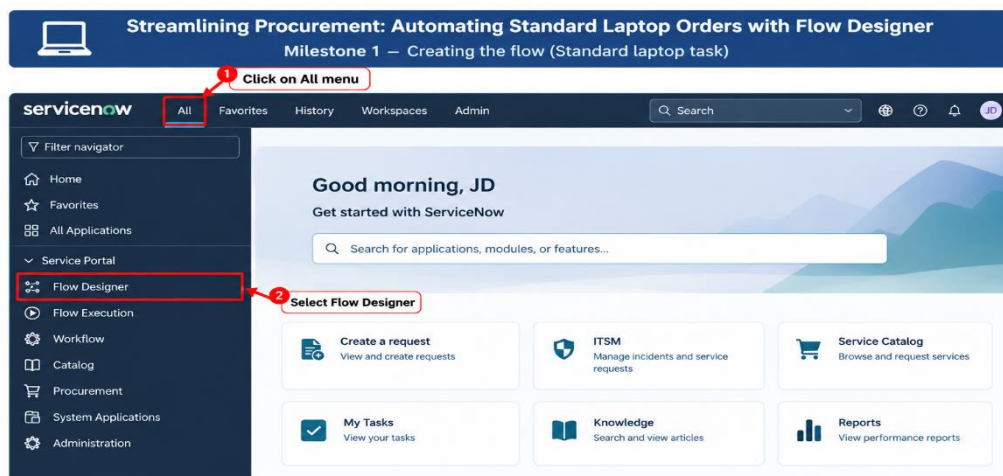


Figure 1: Opening Flow Designer from the All menu

Figure

#### 2: Flow Designer home page (New > Flow)

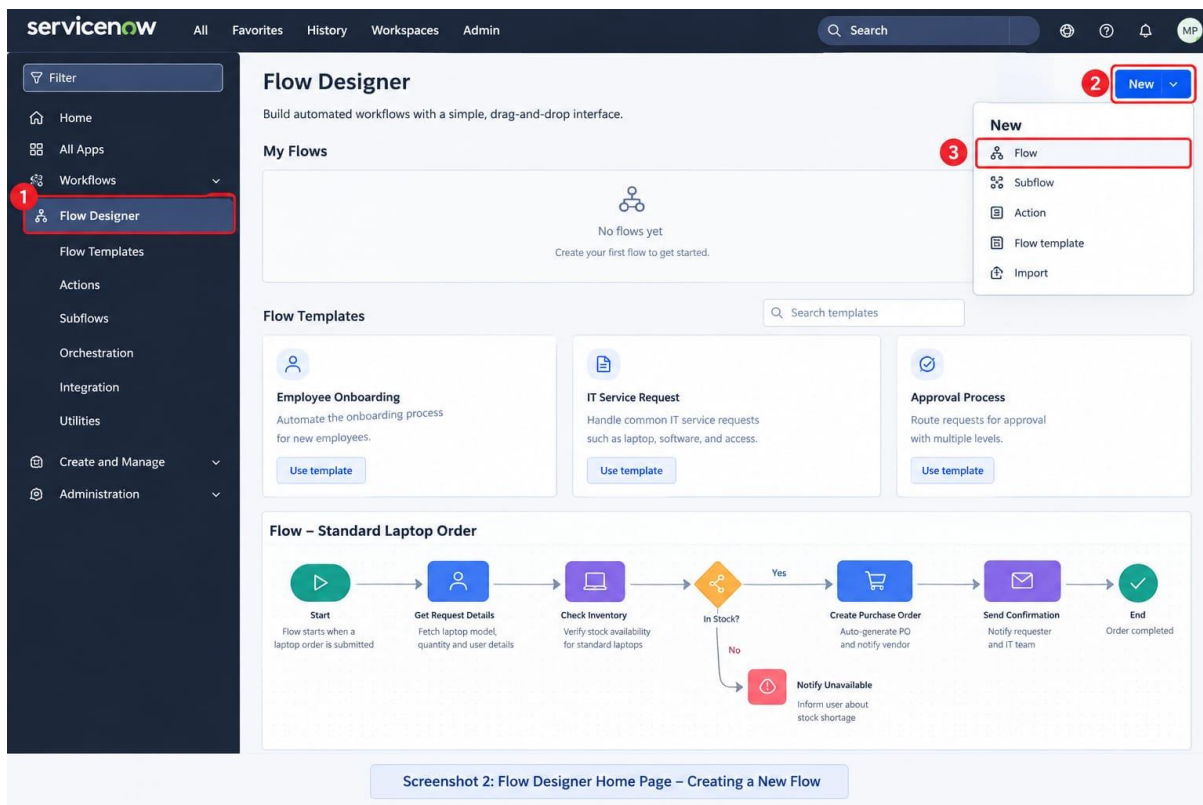


Figure 3: Flow properties (name, application, run user)

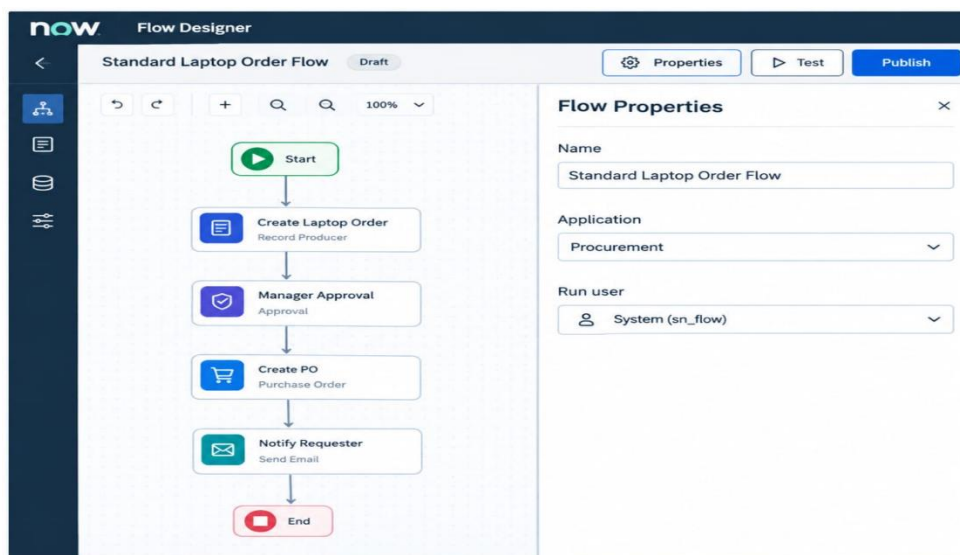


Figure 3: Flow Properties

Figure 4: Adding the Service Catalog trigger

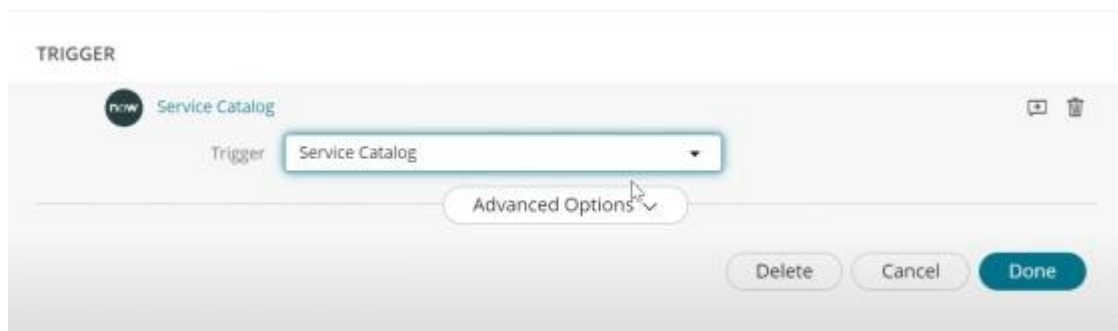


Figure 5: Trigger configured

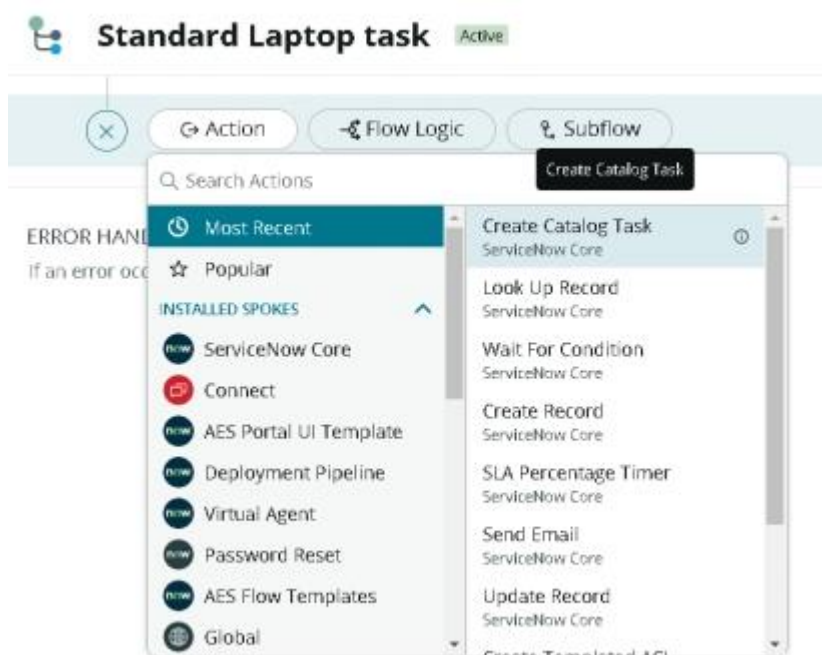


Figure 6: Create Catalog Task action

**Standard Laptop task** Active

View: Top 85 Test De

1 new Create Catalog Task

Action: Create Catalog Task

Table Name: Catalog Task [sc\_task]

Requested item [Requested item]: Trigger ... Requested item ...

Short Description: Laptop needs to Configured

Fields:

- Description: Laptop needs to Configured
- Assignment group: Hardware
- Approval: Approved

+ Add field value

Wait: ☒

Template Catalog: ...

Figure 7: Action field values

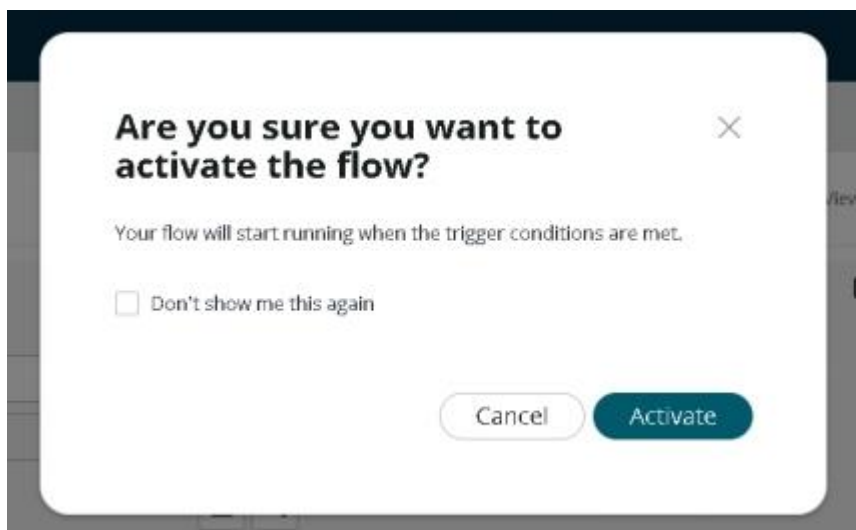


Figure 8: Activating the flow

Milestone 2 – Flow assignment to the Standard Laptop catalog item

| Name                 | Short description                           | Active | Rules | Catalogs        | Category    | Price      | Type | Updated             |
|----------------------|---------------------------------------------|--------|-------|-----------------|-------------|------------|------|---------------------|
| Standard Laptop      | Laptop - CarbonX                            | true   |       | Service Catalog | Hardware    | \$1,100.00 | Item | 2024-04-16 04:54:50 |
| Standard 27" Monitor | Size: 27" Resolution: 1920x1080...          | true   |       | Service Catalog | Peripherals | \$0.00     | Item | 2022-11-20 20:46:33 |
| Standard 22" Monitor | Full HD, IPS Panel with VGA, HDMI, DVI...   | true   |       | Service Catalog | Peripherals | \$200.00   | Item | 2016-05-08 16:49:50 |
| Standard 24" Monitor | TUV-certified Monitor Has Flicker-free S... | true   |       | Service Catalog | Peripherals | \$292.20   | Item | 2022-11-20 20:46:33 |

Figure 9: Maintain Items

Name:  Application:

Catalog:  Active: ☒

Category:  Fulfillment automation level:

State:

Checked out:

Owner:

Item Details | **Process Engine** | Picture | Pricing | Portal Settings

Select the appropriate process engine for the catalog item. Only one engine can be selected.

Flow:

Workflow:

Execution Plan:

Update Copy Try It **Edit in Catalog Builder** Delete

Figure 10: Flow added under Process Engine

### Milestone 3 – Ordering, approval and Catalog Task

Favorites | History | Workspaces | Admin | **Standard Laptop** | Search

Service Catalog > Hardware > Standard Laptop

**Lenovo - Carbon x1**

**x1 Carbon**  
 The x1 Carbon is Lenovo's lightest ThinkPad yet. It provides a QHD display that fights glare and weighs less than three pounds, ideal for most computing tasks, and highly mobile.

**Technical Specs**

- Intel core i5 processor
- 512GB solid state drive (SSD)
- Backlit keyboard

**Optional Software**

- ☐ Adobe Acrobat
- ☐ Adobe Photoshop

Additional software requirements:

**Order this item**  
 Price: \$91,931.2306 + \$8,357.3846 Annually  
 Quantity:   
 Subtotal: \$91,931.2306 + \$8,357.3846 Annually  
 Delivery: **Order Now** 0 Days  
**Order Now**  
 Add to Cart  
 Shopping Cart: Empty

Figure 11: Standard Laptop in Service Catalog

Order Status | Back to Catalog | Continue Shopping | Home

Thank you, your request has been submitted

Order Placed: 2024-04-17 04:07:11  
 Request Number: REQ0010001 ☆  
 Estimated Delivery Date of Complete Order: 2024-04-17

| Description (Includes Annual Charges) | Delivery Date | Stage | Price (ea.)                           | Quantity | Total                                        |
|---------------------------------------|---------------|-------|---------------------------------------|----------|----------------------------------------------|
| Lenovo - Carbon x1                    | 2024-04-17    |       | \$91,931.2306 + \$8,357.3846 Annually | 1        | \$91,931.2306 + \$8,357.3846 Annually        |
| <b>Total</b>                          |               |       |                                       |          | <b>\$91,931.2306 + \$8,357.3846 Annually</b> |

Back to Catalog | Continue Shopping | Home

Figure 12: Order status

The screenshot shows the 'Request - REQ0010001' interface. At the top, there are tabs for 'Favorites', 'History', 'Workspaces', and 'Admin'. Below these is a search bar and a list of actions: 'Discuss', 'Follow', 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. The main section contains input fields for 'Short description' and 'Special instructions'. Below these are buttons for 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. A 'Related Links' section includes 'Show Workflow' and 'Workflow Context'. The 'Requested Items (1)' tab is selected, showing a table with columns: 'State', 'Approver', 'Comments', and 'Created'. The table has one row with the state 'Requested', approver 'Eric Schroeder', and creation time '2024-04-17 04:07:12'.

| State     | Approver       | Comments | Created             |
|-----------|----------------|----------|---------------------|
| Requested | Eric Schroeder |          | 2024-04-17 04:07:12 |

Figure 13: Approvers section of the request

This screenshot shows the 'Approvers' section of the request. A context menu is open over the 'Requested Items (1)' tab, showing options: 'Show Matching', 'Filter Out', 'Copy URL to Clipboard', 'Copy sys\_id', 'Assign Tag', 'Approve', and 'Reject'. The 'Approve' option is highlighted. The table below shows the same data as Figure 12, with one row for 'Requested' by 'Eric Schroeder'.

| State     | Approver       | Comments | Created             |
|-----------|----------------|----------|---------------------|
| Requested | Eric Schroeder |          | 2024-04-17 04:07:12 |

Figure 14: Approving the record

The screenshot shows the 'Requested Item - RITM0010001' interface. At the top, there are tabs for 'Favorites', 'History', 'Workspaces', and 'Admin'. Below these is a search bar and a list of actions: 'Discuss', 'Follow', 'Update', and 'Delete'. The 'Activities' section shows a list of activities for 'System Administrator' with details: 'Impact: 3 - Low', 'Opened by: System Administrator', 'Priority: 4 - Low', and 'State: Open'. The 'Field changes' section shows a change on '2024-04-17 04:07:10'. Below this is a 'Related Links' section with 'Flow Context'. The 'Catalog Tasks (1)' tab is selected, showing a table with columns: 'Number', 'Assignment group', 'Assigned to', 'Short description', 'Actual start', 'Actual end', and 'Order'. The table has one row with 'SCTASK0010001', 'Hardware', '(empty)', 'Laptop needs to be Configured', '2024-04-17 04:11:40', and '(empty)'.

| Number        | Assignment group | Assigned to | Short description             | Actual start        | Actual end | Order |
|---------------|------------------|-------------|-------------------------------|---------------------|------------|-------|
| SCTASK0010001 | Hardware         | (empty)     | Laptop needs to be Configured | 2024-04-17 04:11:40 | (empty)    |       |

*Figure 15: Requested Item record*

The screenshot shows a ServiceNow Catalog Task record for SCTASK0010001. The form includes fields for Number, Assigned to, Configuration item, Active (checkbox), Approval (dropdown), Priority (dropdown), State (dropdown), Request item, and Requested for. The Short description and Description fields both contain the text 'Laptop needs to be configured'. The Worknotes field contains 'Work notes'. The Post button is visible at the bottom right.

*Figure 16: Catalog Task with short description and Hardware group*

## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Catalog Tasks are created automatically, so no configuration step is missed.
- Reduces user waiting time and manual follow-ups.
- Low-code solution that is easy to modify and maintain.
- Better use of Hardware team resources through automatic assignment.
- Every request is traceable from order to task.

### Disadvantages

- Depends on the ServiceNow platform and proper licensing.
- Works only for the Standard Laptop item until extended to other items.
- Incorrect flow configuration can stop task creation, so testing is needed after every change.

## 9. CONCLUSION

By automating the standard laptop procurement process with ServiceNow's Flow Designer, the project effectively addresses inefficiencies and reduces manual overhead. This streamlined workflow ensures timely configuration of laptops, minimises user wait times, and enhances overall productivity within the IT department. The implementation of this automated solution not only optimises resource allocation but also significantly improves the user experience by providing a seamless and efficient procurement process.

## 10. FUTURE SCOPE

- Extend the flow to other catalog items such as laptops of other models, monitors and mobile devices.
- Add e-mail or notification actions to alert the requester when configuration is complete.
- Add conditions for different approval levels based on cost or department.
- Build dashboards and reports on procurement turnaround time.
- Integrate with asset management to update inventory automatically.

## **11. APPENDIX**

Source Code: Not applicable (no-code / low-code flow built in Flow Designer).

Dataset Link: Not applicable.

GitHub & Project Demo Link: [Add your link here]